

STA623 - Bayesian Data Analysis

Practical 2: Priors, conjugate models, posterior summaries & prediction (with solutions)

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Goals. Use conjugate results; check priors by prior predictive simulation; summarise posteriors with point estimates, credible intervals and probabilities; obtain posterior predictive distributions by simulation.

Exercise 1: comparing two groups (General Social Survey)

In the 1990s the General Social Survey recorded the number of children of women aged 40. Women without a college degree: $n_1 = 111$, $\sum_i y_{i,1} = 217$. Women with a degree: $n_2 = 44$, $\sum_i y_{i,2} = 66$. Assume $Y_{i,j} \mid \theta_j \sim_{\text{iid}} \text{Pois}(\theta_j)$ and $\theta_1, \theta_2 \sim_{\text{iid}} \text{Gamma}(2, 1)$.

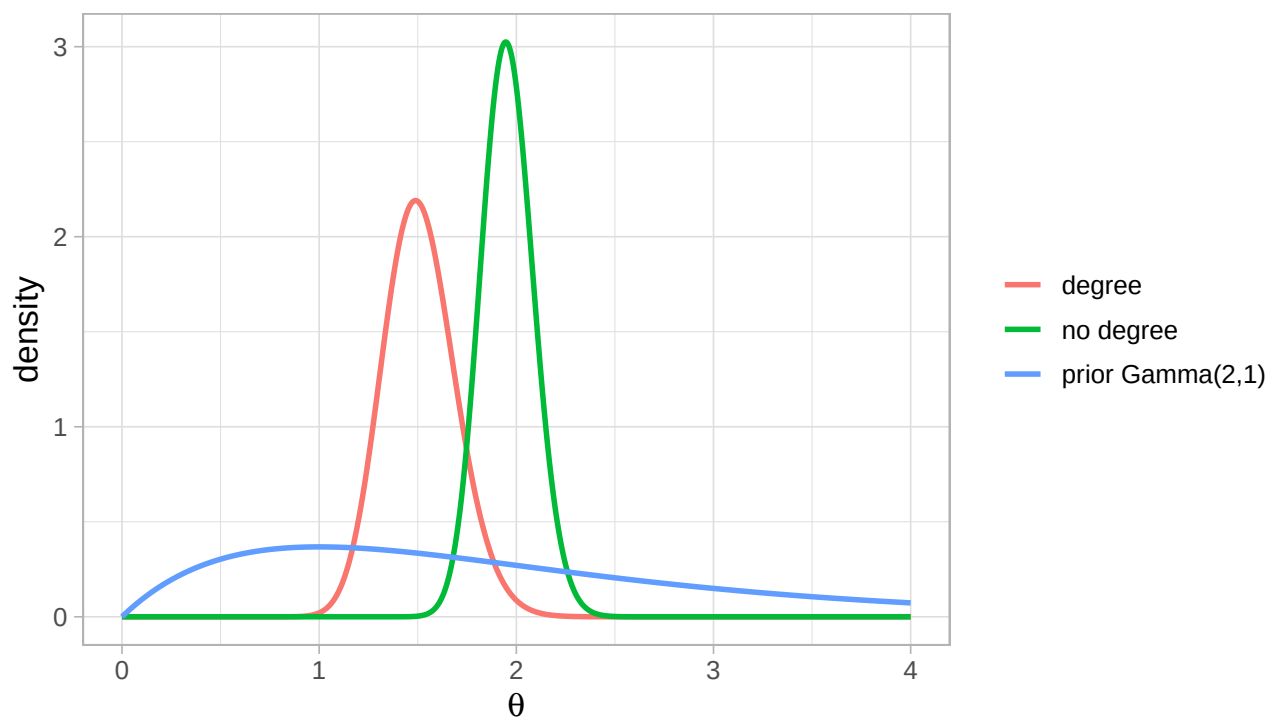
- (a) Write down the two posterior distributions. Plot them on one graph together with the prior.
- (b) Compute posterior means, medians and 95% equal-tailed and HPD intervals for θ_1 and θ_2 .
- (c) Estimate $P(\theta_1 > \theta_2 \mid \mathbf{y})$ and the posterior median and 95% interval of the rate ratio θ_1/θ_2 by simulation.
- (d) Using simulation, obtain the posterior predictive distribution of the number of children of a new woman in each group. Check group 1 against the exact negative binomial result. Estimate $P(\tilde{Y}_1 > \tilde{Y}_2 \mid \mathbf{y})$ and compare with (c). Why is it so much smaller?

Solution

- (a) $\theta_1 \mid \mathbf{y} \sim \text{Gamma}(2 + 217, 1 + 111) = \text{Gamma}(219, 112)$ and $\theta_2 \mid \mathbf{y} \sim \text{Gamma}(68, 45)$.

```
post <- tibble(group = c("no degree", "degree"), a = c(219, 68), b = c(112, 45))

expand_grid(theta = seq(0, 4, length = 800),
             bind_rows(post, tibble(group = "prior Gamma(2,1)", a = 2, b = 1))) |>
  mutate(density = dgamma(theta, a, b)) |>
  ggplot(aes(theta, density, colour = group)) + geom_line(linewidth = 1) +
  labs(x = expression(theta), colour = NULL)
```



(b)

```
post |>
  rowwise() |>
  mutate(mean = a / b, median = qgamma(0.5, a, b),
         ET_lo = qgamma(0.025, a, b), ET_hi = qgamma(0.975, a, b),
         HPD_lo = hdi(qgamma, 0.95, shape = a, rate = b)[1],
         HPD_hi = hdi(qgamma, 0.95, shape = a, rate = b)[2]) |>
  kable(digits = 3)
```

group	a	b	mean	median	ET_lo	ET_hi	HPD_lo	HPD_hi
no degree	219	112	1.955	1.952	1.705	2.223	1.699	2.216
degree	68	45	1.511	1.504	1.173	1.891	1.160	1.875

With this much data the posteriors are nearly symmetric, so equal-tailed and HPD intervals are almost the same.

(c)

```
S <- 1e5
th1 <- rgamma(S, 219, 112); th2 <- rgamma(S, 68, 45)
c(`P(theta1 > theta2)` = mean(th1 > th2), quantile(th1 / th2, c(0.5, 0.025,
  ↪ 0.975)))
```

P(theta1 > theta2)	50%	2.5%	97.5%
0.9726100	1.2980022	0.9946943	1.7181908

(d)

```
y1_new <- rpois(S, th1); y2_new <- rpois(S, th2)

tibble(k = 0:8,
  simulated = map_dbl(k, ~ mean(y1_new == .x)),
  exact_negbin = dnbinom(k, size = 219, prob = 112 / 113)) |>
  kable(digits = 4)
```

k	simulated	exact_negbin
0	0.1425	0.1427
1	0.2776	0.2767
2	0.2697	0.2693
3	0.1756	0.1756
4	0.0852	0.0862
5	0.0347	0.0340
6	0.0110	0.0112
7	0.0027	0.0032
8	0.0008	0.0008

```
c(`P(Ytilde1 > Ytilde2)` = mean(y1_new > y2_new), `P(Ytilde1 = Ytilde2)` =
  ↪ mean(y1_new == y2_new))
```

P(Ytilde1 > Ytilde2)	P(Ytilde1 = Ytilde2)
0.48113	0.21689

We are fairly sure the *average* number of children is higher without a degree (about 97%), but for two *individual* women the one without a degree has more children only about half the time, with ties common. Posterior uncertainty about a population mean is much smaller than the variability between individuals. Confusing the two is a common error in communicating results.

Exercise 2: prior predictive checks

- (a) For the $\text{Gamma}(2, 1)$ prior of Exercise 1, simulate 10^4 values of θ and, for each, the number of children of one woman. What proportion of simulated women have more than 10 children? Does the prior seem reasonable?
- (b) Repeat with the “vague” $\text{Gamma}(0.01, 0.01)$ prior. Comment.
- (c) Suggest a prior for θ that encodes “on average between 1 and 5 children, most likely around 2-3”, and check it by prior predictive simulation.

Solution

```
prior_pred <- function(a, b, S = 1e4) {  
  th <- rgamma(S, a, b); yt <- rpois(S, th)  
  c(`P(y = 0)` = mean(yt == 0), `P(y > 10)` = mean(yt > 10), `99% quantile` =  
    ↪ quantile(yt, 0.99, names = FALSE))  
}  
rbind(`Gamma(2,1)` = prior_pred(2, 1), `Gamma(0.01,0.01)` = prior_pred(0.01,  
  ↪ 0.01),  
      `Gamma(10,4)` = prior_pred(10, 4)) |> kable(digits = 3)
```

	P(y = 0)	P(y > 10)	99% quantile
Gamma(2,1)	0.258	0.003	8
Gamma(0.01,0.01)	0.954	0.018	26
Gamma(10,4)	0.111	0.000	8

- (a) $\text{Gamma}(2, 1)$ has prior mean 2 and allows occasional large families; plausible and not overly restrictive.
- (b) $\text{Gamma}(0.01, 0.01)$ says that almost all women have no children, *and* that a few have implausibly many: it is not “uninformative” about anything we care about. With 111 observations it makes little difference to the posterior, but in small samples it can.
- (c) $\text{Gamma}(10, 4)$ has mean 2.5 and a central 95% range of about 1.2-4.3, matching the description.

```
qgamma(c(0.025, 0.5, 0.975), 10, 4)
```

```
[1] 1.198847 2.417179 4.271201
```

Exercise 3: posterior summaries for a proportion

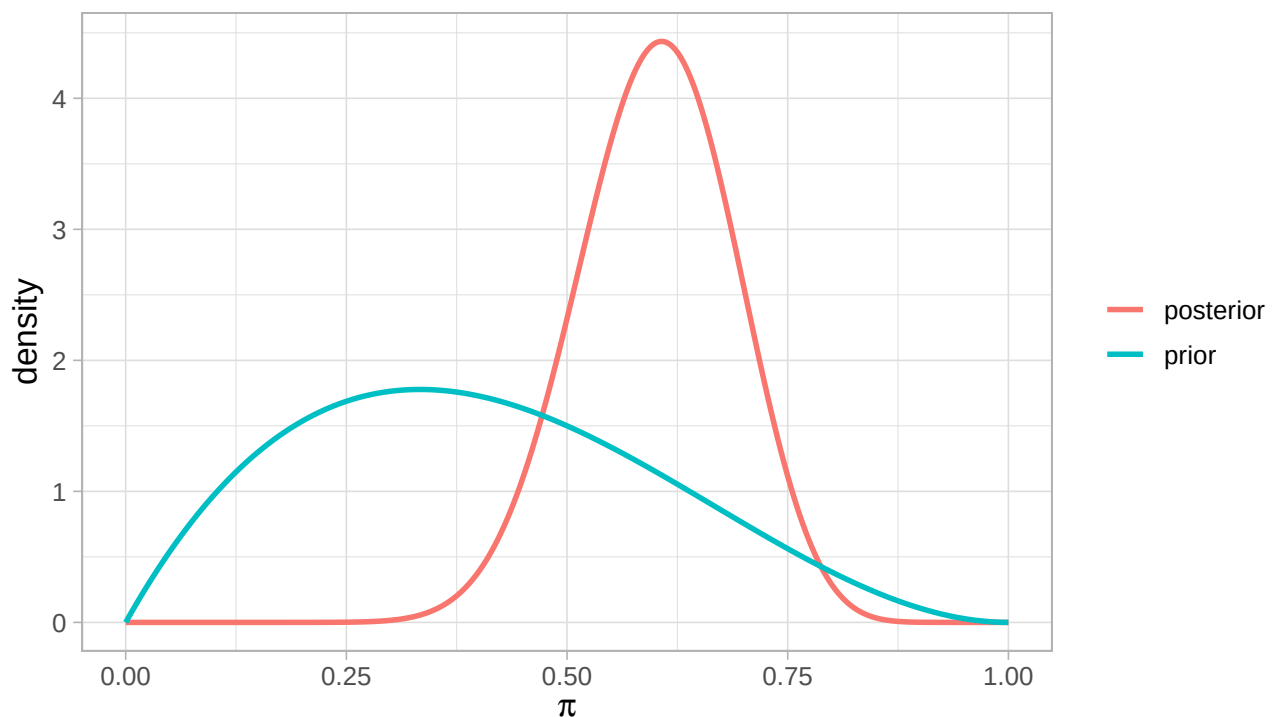
$\pi \sim \text{Beta}(2, 3)$ and $Y_1, \dots, Y_{25} \mid \pi \sim_{\text{iid}} \text{Bernoulli}(\pi)$ with $k = \sum y_i = 16$.

- (a) Find the posterior; plot it against the prior.
- (b) Compute the posterior mean, median and mode. Which loss function justifies each?
- (c) Compute 95% equal-tailed and HPD intervals.
- (d) Compute $P(\pi > 0.5 \mid k)$.
- (e) For $H_1 : \pi \in [0.3, 0.5]$ and $H_2 : \pi \in [0.5, 0.7]$ compute the prior odds, posterior odds and Bayes factor BF_{21} .
- (f) Compute the posterior predictive probability that the next observation is a success, analytically and by simulation.

Solution

- (a) $\text{Beta}(2 + 16, 3 + 9) = \text{Beta}(18, 12)$.

```
a <- 18; b <- 12
tibble(pi = seq(0, 1, length = 500)) |>
  mutate(prior = dbeta(pi, 2, 3), posterior = dbeta(pi, a, b)) |>
  pivot_longer(-pi) |>
  ggplot(aes(pi, value, colour = name)) + geom_line(linewidth = 1) +
  labs(x = expression(pi), y = "density", colour = NULL)
```



(b)-(d)

```
c(mean = a / (a + b),           # squared-error loss
  median = qbeta(0.5, a, b),     # absolute-error loss
  mode = (a - 1) / (a + b - 2),  # 0-1 loss (limit)
  ET = qbeta(c(0.025, 0.975), a, b),
  HPD = hdi(qbeta, 0.95, shape1 = a, shape2 = b),
  `P(pi > 0.5)` = 1 - pbeta(0.5, a, b))
```

mean	median	mode	ET1	ET2	HPD.lower
0.6000000	0.6022507	0.6071429	0.4226046	0.7647598	0.4273464
HPD.upper P(pi > 0.5)					
0.7690367	0.8675346				

(e)

```
pr <- function(lo, hi, a, b) pbeta(hi, a, b) - pbeta(lo, a, b)
prior_odds <- pr(0.5, 0.7, 2, 3) / pr(0.3, 0.5, 2, 3)
post_odds <- pr(0.5, 0.7, a, b) / pr(0.3, 0.5, a, b)
c(prior_odds_H2_vs_H1 = prior_odds, posterior_odds = post_odds, BF21 = post_odds
  ↪ / prior_odds)
```

prior_odds_H2_vs_H1	posterior_odds	BF21
0.6745283	5.5870978	8.2829701

The data multiply the odds in favour of H_2 by a factor of about 8.

(f) $P(\tilde{Y} = 1 \mid k) = E[\pi \mid k] = 18/30 = 0.6$.

```
mean(rbinom(1e5, 1, rbeta(1e5, a, b)))
```

```
[1] 0.59979
```

Exercise 4: prior sensitivity and elicitation

For the data of Exercise 3, a clinician tells you that from experience “the proportion is probably around 40%, and I’d be as confident about that as if I’d seen 20 patients”.

- (a) Turn this into a Beta prior.
- (b) Compare posterior mean, 95% equal-tailed interval and $P(\pi > 0.5 \mid k)$ under: Beta(1,1), Beta($\frac{1}{2}, \frac{1}{2}$), Beta(2,3), the clinician’s prior, and a very strong version of it (worth 200 patients). Summarise in a table and a plot.
- (c) Write two sentences you could put in a report about the sensitivity of the conclusions.

Solution

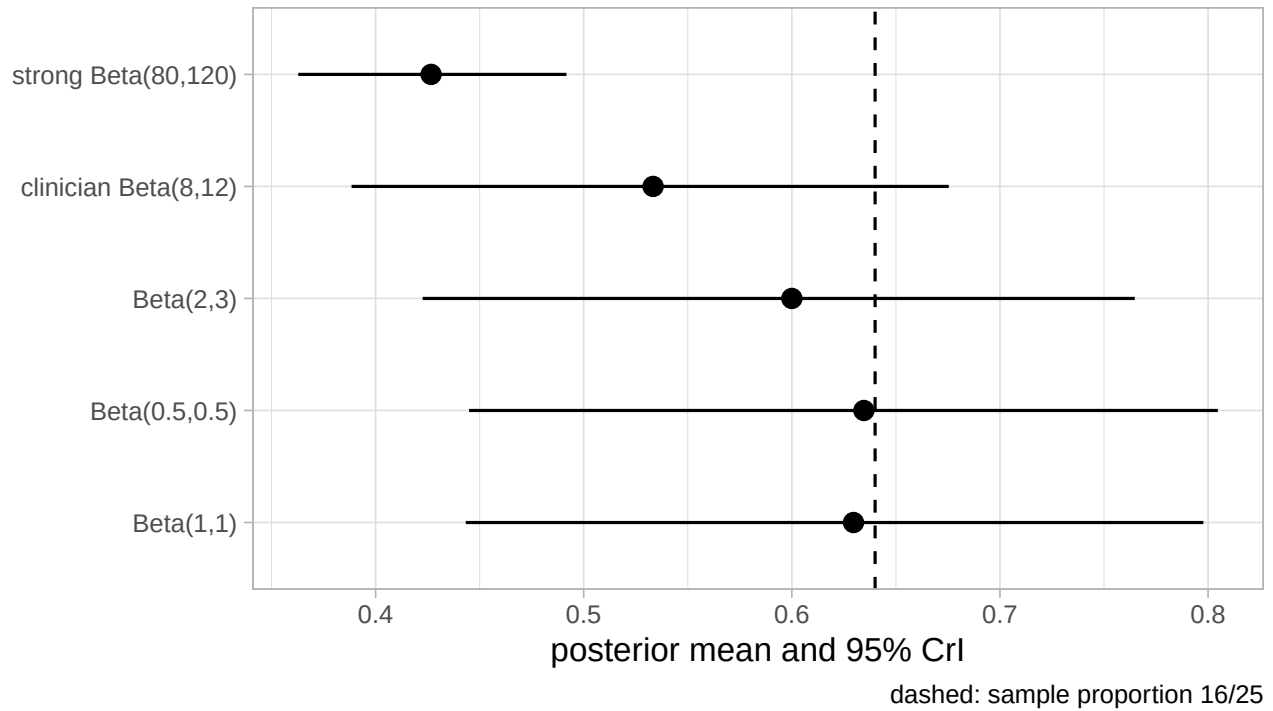
- (a) Prior mean $m = 0.4$, prior sample size $n_0 = 20$: $\text{Beta}(mn_0, (1 - m)n_0) = \text{Beta}(8, 12)$.

- (b)

```
priors <- tribble(~prior, ~a0, ~b0,
  "Beta(1,1)", 1, 1, "Beta(0.5,0.5)", 0.5, 0.5, "Beta(2,3)", 2, 3,
  "clinician Beta(8,12)", 8, 12, "strong Beta(80,120)", 80, 120) |>
mutate(a = a0 + 16, b = b0 + 9,
  mean = a / (a + b), lo = qbeta(0.025, a, b), hi = qbeta(0.975, a, b),
  `P(pi > 0.5)` = 1 - pbeta(0.5, a, b))
kable(priors, digits = 3)
```

prior	a0	b0	a	b	mean	lo	hi	P(pi > 0.5)
Beta(1,1)	1.0	1.0	17.0	10.0	0.630	0.443	0.798	0.916
Beta(0.5,0.5)	0.5	0.5	16.5	9.5	0.635	0.445	0.805	0.920
Beta(2,3)	2.0	3.0	18.0	12.0	0.600	0.423	0.765	0.868
clinician Beta(8,12)	8.0	12.0	24.0	21.0	0.533	0.388	0.675	0.674
strong Beta(80,120)	80.0	120.0	96.0	129.0	0.427	0.363	0.492	0.014

```
ggplot(priors, aes(mean, fct_inorder(prior))) +
  geom_pointrange(aes(xmin = lo, xmax = hi)) +
  geom_vline(xintercept = 16 / 25, linetype = 2) +
  labs(x = "posterior mean and 95% CrI", y = NULL, caption = "dashed: sample
  ↪ proportion 16/25")
```



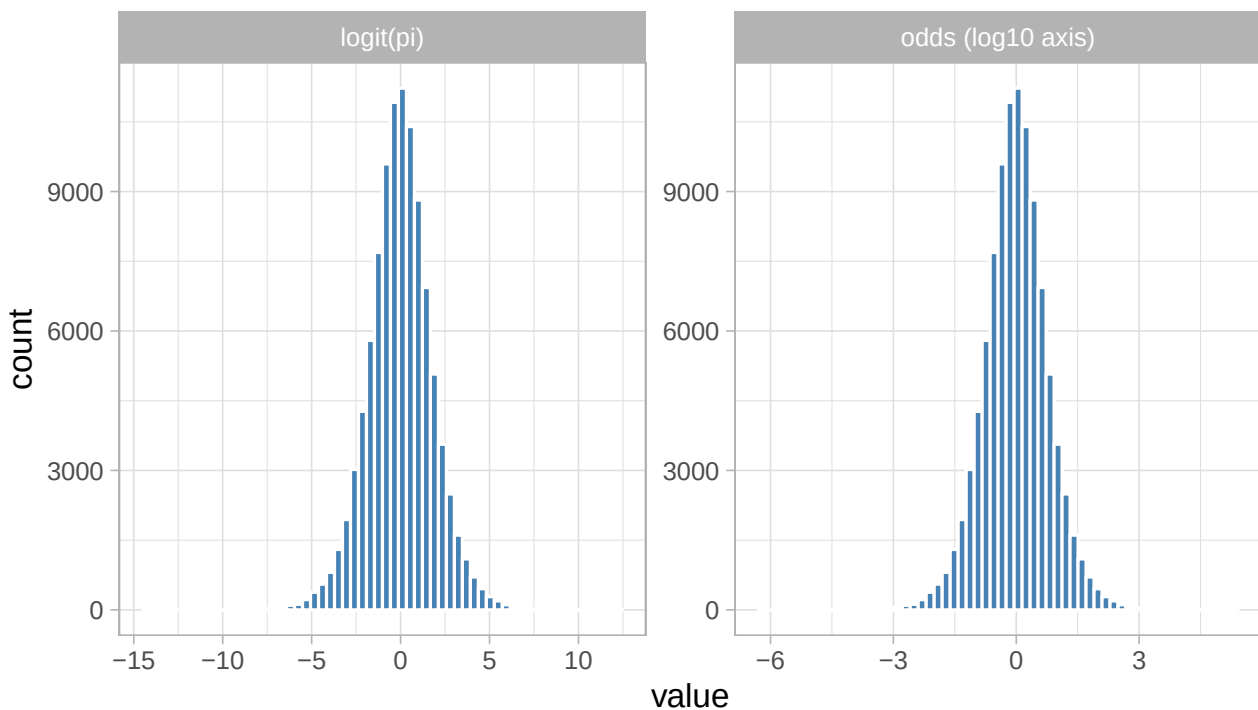
- (c) Under a range of vague and weakly informative priors the posterior probability that $\pi > 0.5$ is about 0.87-0.92; incorporating the clinician's prior (worth 20 patients, centred on 40%) reduces it to about 0.67. A prior as strong as 200 patients would dominate the 25 observations and reverse the conclusion ($P(\pi > 0.5) \approx 0.01$), so it should only be used if such evidence genuinely exists.

Exercise 5: priors on transformed scales

- (a) Simulate 10^5 draws of $\pi \sim \text{Uniform}(0, 1)$ and plot the implied distribution of $\text{logit}(\pi)$ and of the odds $\pi/(1 - \pi)$.
- (b) In a logistic regression, a log odds ratio β is often given the prior $\mathcal{N}(0, 10^2)$. What does this imply for the odds ratio e^β ? Compute $P(e^\beta > 100) + P(e^\beta < 0.01)$. Compare with $\mathcal{N}(0, 1)$.

Solution

```
u <- runif(1e5)
tibble(`logit(pi)` = qlogis(u), `odds (log10 axis)` = log10(u / (1 - u))) |>
  pivot_longer(everything()) |>
  ggplot(aes(value)) + geom_histogram(bins = 60, fill = "steelblue", colour =
    ↪ "white") +
  facet_wrap(~name, scales = "free")
```



A flat prior on π is a logistic distribution on the log-odds, concentrated between about -5 and 5.

```
beta10 <- rnorm(1e5, 0, 10); beta1 <- rnorm(1e5, 0, 1)
c(`N(0,10^2)` = mean(exp(beta10) > 100 | exp(beta10) < 0.01),
  `N(0,1)`     = mean(exp(beta1) > 100 | exp(beta1) < 0.01))
```

$\mathcal{N}(0, 10^2)$	$\mathcal{N}(0, 1)$
0.64376	0.00001

$\mathcal{N}(0, 10^2)$ puts about 65% of its mass on odds ratios above 100 or below 1/100, which is implausible for almost any epidemiological exposure. $\mathcal{N}(0, 1)$ puts essentially none there. “Vague” on one scale can be strongly (and wrongly) informative on the scale that matters.

Exercise 6

- (a) Derive the Jeffreys prior for a Poisson sampling model with n observations. Is it proper? Is the posterior proper?
- (b) Find the conjugate prior for (i) a Geometric(π) sampling model, $p(y | \pi) = (1-\pi)^y \pi$, $y = 0, 1, \dots$; (ii) an Exponential(λ) sampling model.
- (c) Show that the posterior predictive distribution for the Gamma-Poisson model is negative binomial.

Solution

- (a) $\log p(\mathbf{y} | \lambda) = \text{const} + \sum y_i \log \lambda - n\lambda$, so $-\partial^2 / \partial \lambda^2 = \sum y_i / \lambda^2$, with expectation $I(\lambda) = n/\lambda$. Hence $p_J(\lambda) \propto \lambda^{-1/2}$, improper ($\int_0^\infty \lambda^{-1/2} d\lambda = \infty$). The posterior $\propto \lambda^{\sum y_i + 1/2 - 1} e^{-n\lambda}$ is Gamma($\sum y_i + \frac{1}{2}, n$), proper for $n \geq 1$.
- (b) (i) Likelihood $\propto \pi^n (1-\pi)^{\sum y_i}$: conjugate prior Beta(a, b), posterior Beta($a+n, b+\sum y_i$). (ii) Likelihood $\lambda^n e^{-\lambda \sum y_i}$: conjugate prior Gamma(a, b), posterior Gamma($a+n, b+\sum y_i$).
- (c) See Appendix B.2 of the lecture notes.